

Muhammad Danish

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Education

University of New Mexico

- Ph.D. in Computer Science
- M.S. in Computer Science
- B.S. in Computer Science; Minor in Honors Interdisciplinary Liberal Arts

Albuquerque, NM

Aug 2025 – May 2028

Aug 2025 – Jul 2026

Aug 2021 – May 2025

Research & Publications

1. Danish, M., et al. “The Tragedy of Convenience: Cascading User-Data Leakage from SMS-Delivered URLs.” *ACM Conference on Computer and Communications Security (CCS)*, 2026.

Led a large-scale study of 33M+ SMS messages and 322K URLs; built an end-to-end LLM-assisted pipeline to identify 170+ PII-exposing services; Performed token-entropy analysis, bounded enumeration, authentication testing, API and WebSocket analysis, and coordinated disclosure with remediations protecting 120M+ users.

2. Sobrados, E., Danish, M., et al. “Exposing Probe-Resistant VPNs via Protocol Coexistence.” *IEEE Symposium on Security and Privacy (SP)*, 2027.

Developed an Internet-measurement pipeline that reverse-engineered commercial VPN applications and enumerated 72,887 servers; found that 83.4% were fingerprintable because probe-resistant protocols coexisted with detectable legacy or misconfigured protocols.

3. Danish, M., et al. “Measuring View-Based Monetization Fraud Across Content-Sharing Platforms.” Under review at the *USENIX Security*, 2027.

Led a cross-platform study of view inflation across a diverse set of monetizable content-sharing services (e.g., Spotify, Dailymotion, Ameblo); 9 of 10 platforms counted more than 80% of attempted views and attacks on five platforms persisted & scaled using consumer-grade infrastructure;

4. Danish, M., et al. “Security Under Compression: Evaluating the Security and Privacy Impacts of Model Compression.” Under review at *USENIX Security*, 2027.

Developing a systematic cross-attack evaluation framework to quantify how model compression techniques, including quantization, pruning, knowledge distillation, and low-rank factorization, affect security, privacy, and adversarial robustness against a comprehensive set of attacks.

Experience

- **Research Assistant (PI and Reference: [Afsah Anwar](#)), University of New Mexico** Dec 2023 – Present
 - Conducted large-scale AI, security & privacy research, resulting in multiple publications.
 - Built end-to-end pipelines and reverse-engineered AI/web/mobile systems to uncover vulnerabilities.
 - Characterized security & privacy issues in multiple systems (e.g., web, AI/ML) and proposed mitigation strategies.
- **Student Programming Analyst, University of New Mexico** Oct 2022 – Aug 2025
 - Automated cross-department administrative workflows using Microsoft Power Platform, Power Automate, and Boomi.
 - Built regression-test suites and integrated them into Azure DevOps CI/CD pipelines, reducing testing time by ~50%.
 - Delivered web applications and system integrations across multiple departments, supporting 20,000+ students.
- **Computer Science Teaching Assistant, University of New Mexico**
 - Led weekly programming, networking, and security labs for 40+ students.
 - Developed exercises, graded assignments and examinations, and provided individualized technical feedback.
 - Fostered a collaborative learning environment via group discussions, one-on-one sessions, and [YouTube](#).
 - **Courses:** Introduction to Cybersecurity, Computer Networks, and Intro/Intermediate Programming

Awards & Skills

- **Awards:** [CRA Outstanding Undergraduate Researcher](#) (North America), NMEF Scholar (New Mexico), [Bug Bounty](#)
- **Languages:** Python (PyTorch and TensorFlow), Bash, PowerShell, Java, JavaScript, SQL, MATLAB
- **Technical:** model compression, adversarial ML, LLMs, agentic AI, web & API security, reverse engineering
- **Tools:** Linux/Windows, Playwright, Docker, NMap, Wireshark, Burp Suite, Kibana, Metasploit, Azure, Git

Volunteering

- **Peer Review:** AEC member at top conferences e.g., IEEE S&P ([Award](#)), ACM CCS, NDSS, PETS ([Award](#))
- **Stanford Code in Place (Section Leader):** Mentored diverse students in Python and CS fundamentals.